

Answer Key & Teacher Notes

Session 1 – Surface Area of Pyramids • Session 2 – Lateral Area of Pyramids

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Surface Area of Pyramids

TALK PROMPTS

- Why do you use the slant height instead of the pyramid's straight-up height?
- Which measure belongs to the square, and which one belongs to the triangles?

WATCH FOR

- Adding the base and only one triangle.
- Using the pyramid's straight-up height instead of the slant height.
- Forgetting the $\frac{1}{2}$ in each triangle's area.

PARTNER TASK

Pyramid Tent. A tent has a square base 6 ft on each side and a slant height of 5 ft. One partner finds the base (36 ft^2), the other one triangle (15 ft^2). Together decide how many triangles to add ($36 + 60 = 96 \text{ ft}^2$), then decide what changes if the floor is open ground (60 ft^2).

SESSION 2 • TEACHER NOTES

Lateral Area of Pyramids

TALK PROMPTS

- When should you include the base in a total, and when should you leave it out?
- What words in a problem tell you to leave the base out?

WATCH FOR

- Adding the base to a lateral area when the base is not covered.
- Using 4 slanted faces for a pyramid with a triangle base (it has 3).
- Forgetting the $\frac{1}{2}$ in each triangle's area.

PARTNER TASK

Two Pyramids. Both have a base 9 in on each side and a slant height of 12 in. Partner A paints only the slanted faces (216 in^2); Partner B paints every face (297 in^2). Together explain why the answers differ by exactly 81 in^2 .

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.G.A.4	1	1	1	1	2	2	1	3	3	1	1
S1 • Form B	6.G.A.4	1	1	1	1	2	2	1	3	3	1	1
S2 • Form A	6.G.A.4	1	2	1	1	1	2	2	3	3	1	1
S2 • Form B	6.G.A.4	1	2	1	1	1	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

slant height: The height of a side triangle, measured along its face.

lateral face: A side **triangle** of a pyramid.

apex: The **top** point where the side faces meet.

- SA of a pyramid = base area + area of all the **triangular** faces.
- A square pyramid has **4** matching triangles.
- Use the **slant** height, not the straight-up height, for the triangles.

Example: Base: $10 \times 10 = 100 \text{ cm}^2$. One triangle: $\frac{1}{2} \times 10 \times 7 = 35 \text{ cm}^2$. Four triangles: $4 \times 35 = 140 \text{ cm}^2$. Answer: SA = $100 + 140 = 240 \text{ cm}^2$

YOU TRY

1. $16 + 4 \times (\frac{1}{2} \times 4 \times 6) = 16 + 48 = 64 \text{ m}^2$
2. a. $\frac{1}{2} \times 6 \times 5 = 15 \text{ in}^2$ b. 36 in^2

EXIT TICKET

1. a. 81 cm^2 b. $\frac{1}{2} \times 9 \times 4 = 18 \text{ cm}^2$
2. $36 + 4 \times (\frac{1}{2} \times 6 \times 7) = 36 + 84 = 120 \text{ in}^2$
3. **A** (the slant height)

PRACTICE FORM A

1. a. $64 + 4 \times 40 = 224 \text{ m}^2$ b. $25 + 4 \times 20 = 105 \text{ ft}^2$ c. $4 + 4 \times 3 = 16 \text{ cm}^2$ d. $49 + 4 \times 35 = 189 \text{ in}^2$
2. a. $16 \times 16 = 256 \text{ cm}^2$ b. $\frac{1}{2} \times 16 \times 15 = 120 \text{ cm}^2$ c. $256 + 4 \times 120 = 736 \text{ cm}^2$
3. $64 + 4 \times (\frac{1}{2} \times 8 \times 11) = 64 + 176 = 240 \text{ in}^2$
4. $144 + 4 \times (\frac{1}{2} \times 12 \times 9) = 144 + 216 = 360 \text{ cm}^2$
5. Base $20 \times 20 = 400 \text{ ft}^2$; triangles $4 \times (\frac{1}{2} \times 20 \times 26) = 1,040 \text{ ft}^2$; SA = $1,440 \text{ ft}^2$
6. No. 117 cm^2 is one triangle. SA = $324 + 4 \times 117 = 792 \text{ cm}^2$.
7. $256 + 4 \times (\frac{1}{2} \times 16 \times 10) = 256 + 320 = 576 \text{ in}^2$
8. a. base 196 in^2 ; one triangle $\frac{1}{2} \times 14 \times 15 = 105 \text{ in}^2$ b. $196 + 4 \times 105 = 616 \text{ in}^2$ c. $420 \times 0.04 = \$16.80$
9. Ethan added only one triangle. A square pyramid has 4 triangles: $4 \times 187 = 748$. SA = $484 + 748 = 1,232 \text{ in}^2$.
10. **B** (4)
11. **D** (105 cm^2)

Self-check word PYRAMIDS

PRACTICE FORM B

1. a. $36 + 4 \times 24 = 132 \text{ m}^2$ b. $16 + 4 \times 14 = 72 \text{ ft}^2$ c. $9 + 4 \times 7 \frac{1}{2} = 39 \text{ cm}^2$ d. $81 + 4 \times 45 = 261 \text{ in}^2$
2. a. $14 \times 14 = 196 \text{ cm}^2$ b. $\frac{1}{2} \times 14 \times 12 = 84 \text{ cm}^2$ c. $196 + 4 \times 84 = 532 \text{ cm}^2$
3. $100 + 4 \times (\frac{1}{2} \times 10 \times 13) = 100 + 260 = 360 \text{ in}^2$
4. $64 + 4 \times (\frac{1}{2} \times 8 \times 9) = 64 + 144 = 208 \text{ cm}^2$
5. Base $24 \times 24 = 576 \text{ ft}^2$; triangles $4 \times (\frac{1}{2} \times 24 \times 20) = 960 \text{ ft}^2$; SA = $1,536 \text{ ft}^2$
6. No. 88 cm^2 is one triangle. SA = $256 + 4 \times 88 = 608 \text{ cm}^2$.
7. $144 + 4 \times (\frac{1}{2} \times 12 \times 14) = 144 + 336 = 480 \text{ in}^2$
8. a. base 100 in^2 ; one triangle $\frac{1}{2} \times 10 \times 12 = 60 \text{ in}^2$ b. $100 + 4 \times 60 = 340 \text{ in}^2$ c. $240 \times 0.05 = \$12$
9. Mia added only one triangle. A square pyramid has 4 triangles: $4 \times 150 = 600$. SA = $400 + 600 = 1,000 \text{ in}^2$.
10. **B** (5)
11. **C** (96 cm^2)

Self-check word MONUMENT

GUIDED NOTES

lateral area: The area of the slanted faces **only**.

surface area: The area of **every** face, base included.

lateral face: A **slanted** side triangle of a pyramid.

- Lateral area = the area of all the **slanted** faces.
- Surface area = lateral area + **base** area.
- Read the problem to decide whether the base is **covered**.

Example: Cover the 4 slanted triangles. Leave out the **base**. One triangle: $\frac{1}{2} \times 4 \times 6 = 12 \text{ m}^2$. Four triangles: $4 \times 12 = 48 \text{ m}^2$. Answer: lateral area = **48 m²**

YOU TRY

- $4 \times (\frac{1}{2} \times 6 \times 7) = 4 \times 21 = 84 \text{ ft}^2$
- a. $90 - 25 = 65 \text{ m}^2$ b. $3 \times 12 = 36 \text{ in}^2$

EXIT TICKET

- a. 60 cm^2 b. $100 - 36 = 64 \text{ in}^2$
- $4 \times (\frac{1}{2} \times 10 \times 6) = 4 \times 30 = 120 \text{ m}^2$
- A** (lateral area)

PRACTICE FORM A

- a. $4 \times 25 = 100 \text{ cm}^2$ b. $150 - 36 = 114 \text{ in}^2$ c. $4 \times (\frac{1}{2} \times 7 \times 9) = 126 \text{ m}^2$ d. $4 \times 55 = 220 \text{ in}^2$
- a. 3 b. $\frac{1}{2} \times 24 \times 14 = 168 \text{ in}^2$ c. $3 \times 168 = 504 \text{ in}^2$
- a. $\frac{1}{2} \times 10 \times 9 = 45 \text{ ft}^2$ b. $4 \times 45 = 180 \text{ ft}^2$ c. $180 + 100 = 280 \text{ ft}^2$
- $4 \times (\frac{1}{2} \times 14 \times 8) = 4 \times 56 = 224 \text{ ft}^2$
- $4 \times (\frac{1}{2} \times 22 \times 17) = 4 \times 187 = 748 \text{ m}^2$
- $676 + 4 \times (\frac{1}{2} \times 26 \times 25) = 676 + 1,300 = 1,976 \text{ in}^2$
- No. Kai forgot the $\frac{1}{2}$. One face is $\frac{1}{2} \times 30 \times 21 = 315 \text{ cm}^2$, so the lateral area is $4 \times 315 = 1,260 \text{ cm}^2$.
- a. $4 \times (\frac{1}{2} \times 16 \times 10) = 320 \text{ ft}^2$ b. $320 \times 3 = \$960$ c. $320 + 256 = 576 \text{ ft}^2$; the floor is not covered, so the lateral area is the canvas needed.
- Rosa added the base (196 cm^2), but the base is not painted. The painted area is the lateral area only: $4 \times (\frac{1}{2} \times 14 \times 8) = 224 \text{ cm}^2$.
- A** (base)
- A** (3)

Self-check word SHINGLES

PRACTICE FORM B

- a. $4 \times 30 = 120 \text{ cm}^2$ b. $180 - 64 = 116 \text{ in}^2$ c. $4 \times (\frac{1}{2} \times 5 \times 8) = 80 \text{ m}^2$ d. $4 \times 54 = 216 \text{ in}^2$
- a. 3 b. $\frac{1}{2} \times 18 \times 10 = 90 \text{ in}^2$ c. $3 \times 90 = 270 \text{ in}^2$
- a. $\frac{1}{2} \times 12 \times 10 = 60 \text{ ft}^2$ b. $4 \times 60 = 240 \text{ ft}^2$ c. $240 + 144 = 384 \text{ ft}^2$
- $4 \times (\frac{1}{2} \times 16 \times 9) = 4 \times 72 = 288 \text{ ft}^2$
- $4 \times (\frac{1}{2} \times 18 \times 15) = 4 \times 135 = 540 \text{ m}^2$
- $400 + 4 \times (\frac{1}{2} \times 20 \times 24) = 400 + 960 = 1,360 \text{ in}^2$
- No. Luis forgot the $\frac{1}{2}$. One face is $\frac{1}{2} \times 24 \times 15 = 180 \text{ cm}^2$, so the lateral area is $4 \times 180 = 720 \text{ cm}^2$.
- a. $4 \times (\frac{1}{2} \times 12 \times 8) = 192 \text{ ft}^2$ b. $192 \times 4 = \$768$ c. $192 + 144 = 336 \text{ ft}^2$; the floor is not covered, so the lateral area is the canvas needed.
- Grace added the base (144 cm^2), but the base is not painted. The painted area is the lateral area only: $4 \times (\frac{1}{2} \times 12 \times 7) = 168 \text{ cm}^2$.
- A** (base area)
- B** (4)

Self-check word COVERAGE

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • SURFACE AREA OF PYRAMIDS

1. $36 + 4 \times (\frac{1}{2} \times 6 \times 9) = 36 + 108 = 144 \text{ m}^2$
2. a. 144 cm^2 b. $\frac{1}{2} \times 12 \times 10 = 60 \text{ cm}^2$ SA = $144 + 240 = 384 \text{ cm}^2$
3. a. $16 + 4 \times (\frac{1}{2} \times 4 \times 5) = 56 \text{ in}^2$ b. $100 + 4 \times (\frac{1}{2} \times 10 \times 12) = 340 \text{ cm}^2$
4. $81 + 4 \times (\frac{1}{2} \times 9 \times 11) = 81 + 198 = 279 \text{ in}^2$
5. He used the pyramid's height (4 ft) instead of the slant height (5 ft). SA = $36 + 4 \times (\frac{1}{2} \times 6 \times 5) = 36 + 60 = 96 \text{ ft}^2$.

SHEET 2 • LATERAL AREA OF PYRAMIDS

1. $4 \times (\frac{1}{2} \times 8 \times 6) = 96 \text{ ft}^2$
2. a. 136 in^2 b. 72 cm^2 c. $4 \times (\frac{1}{2} \times 6 \times 10) = 120 \text{ m}^2$ d. $4 \times (\frac{1}{2} \times 3 \times 4) = 24 \text{ ft}^2$
3. Lateral area $4 \times (\frac{1}{2} \times 14 \times 10) = 280 \text{ ft}^2$; $280 \div 35 = 8$ bundles
4. One face: $160 \div 4 = 40 \text{ in}^2$; $\frac{1}{2} \times 8 \times \text{slant height} = 40$, so slant height = 10 in
5. No. $85 \text{ ft}^2 = 25 + 60$ includes the floor. With no floor, only the lateral area is needed: $4 \times (\frac{1}{2} \times 5 \times 6) = 60 \text{ ft}^2$.
6. **B** (56 cm^2)
7. **C** (36 m^2)